

spinalSynth MIDI PICO2

A standalone, real-time hardware USB MIDI-to-UART bridge implemented on the **Raspberry Pi Pico 2 (RP2350)** using the official Pico SDK and TinyUSB.

The Pico 2 acts as a **USB MIDI Host** to interface with multiple physical MIDI devices (via a USB Hub) and stream UART register update frames directly to the spinalSynth hardware.

Hardware Configuration & Pinout

1. Connecting MIDI Controllers

- Because the Pico 2 has a single USB port, you must connect a standard **USB Hub** to the Pico 2's USB-C port using an OTG adapter cable.
- Plug your MIDI controllers (e.g. keyboard, CC controller) directly into the USB Hub. The Pico 2 will supply host power and enumerate all controllers concurrently.

2. Wiring to spinalSynth (UART) & Debug Console

Connect the Pico 2 UART TX pins directly to the spinalSynth RX pins (3.3V logic levels). A secondary UART port is used for debugging console logs to a PC:

Pico 2 Pin	Function	Target / Connection
GPIO 0 (Pin 1)	UART0 TX	spinalSynth RX (Command Input)
GPIO 1 (Pin 2)	UART0 RX	spinalSynth TX (Optional Command Output)
GPIO 4 (Pin 6)	UART1 TX	Debug Console RX (115200 Baud serial monitor)
GPIO 5 (Pin 7)	UART1 RX	Debug Console TX (Optional serial command input)
GND	Ground	Common Ground

Note: Since the onboard USB-C port is configured in USB Host mode to talk to the MIDI Hub/devices, debugging stdout cannot run over USB CDC. It is redirected to UART1 (Pins GP4/GP5) at 115200 baud.

MIDI to Register Mapping

All parameters are mapped sequentially across CC 1 to 12:

MIDI Event	Target Register	Address	Description
Note ON	DDS Frequency / Gate	0x00-0x02 / 0x45	Set 24-bit DDS tuning word; writes 0x01 to ENV_GATE
Note OFF	Gate	0x45	Writes 0x00 to ENV_GATE
CC 1	PWM_WIDTH	0x04	Duty cycle for PWM (scaled 0-127 $\rightarrow$ 0-255)
CC 2	WAVE_SEL	0x03	Waveform selection (maps onto states 0 to 5)
CC 3	ENV_ATTACK	0x41	Attack time duration
CC 4	ENV_DECAY	0x42	Decay time duration
CC 5	ENV_SUSTAIN	0x43	Sustain Level
CC 6	ENV_RELEASE	0x44	Release time duration
CC 7	ENV_CTRL [2]	0x40	Loop Mode (ON if $\geq$ 64, OFF if < 64)
CC 8	ENV_CTRL [5:4]	0x40	Curve Select (0=Lin, 1=Exp, 2=Log, 3=S-Curve)
CC 9	FILTER_ENABLE	0x50	Filter Enable (ON if $\geq$ 64, OFF if < 64)
CC 10	FILTER_MODE	0x51	Response Mode (LP if 0-42, BP if 43-85, HP if 86-127)
CC 11	FILTER_CUTOFF	0x52	Cutoff Frequency (scaled 0-127 $\rightarrow$ 0-255)
CC 12	FILTER_RESONANCE	0x53	Resonance / Feedback (scaled 0-127 $\rightarrow$ 0-255)

Technical Details & USB Host Stack

- Native USB Controller (Port 0):** This project runs on the native hardware USB controller of the RP2350, utilizing standard differential signals. This keeps the implementation clean and avoids the complexity, core dedication, clock-scaling, and resource consumption of bit-banged PIO solutions.
- TinyUSB Class Driver Callback:** In Pico SDK 2.x, the MIDI Host class driver is not registered automatically. To bind and mount the USB MIDI device interface when plugged in, the application implements the weak callback `usbh_app_driver_get_cb()` to return the MIDI class driver:

```
usbh_class_driver_t const* usbh_app_driver_get_cb(uint8_t* driver_count) {
    static usbh_class_driver_t const midi_driver = {
        .name = "MIDI",
        .init = midih_init,
        .deinit = midih_deinit,
        .open = midih_open,
        .set_config = midih_set_config,
        .xfer_cb = midih_xfer_cb,
        .close = midih_close
    };
    *driver_count = 1;
    return &midi_driver;
}
```

Implementing this callback registers the driver and resolves device-recognition/mounting issues.

Build Instructions

Using the official Raspberry Pi Pico VS Code Extension:

1. Open the `/spinalSynth_MIDI_PICO2` project folder in VS Code.
2. The extension will automatically configure the CMake environment for the **RP2350** processor.
3. Click the **Compile** button in the VS Code status bar (or run **CMake: Build** from the command palette).
4. Put your Pico 2 in bootsel mode, and copy the compiled `.uf2` file to the drive (or click **Flash** with a debugger connected).